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Cooperative decision-making algorithm with efficient convergence for UCAV formation in beyond-visual-range air combat based on multi-agent reinforcement learning



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Advantage highlight;
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Abstract Highly intelligent Unmanned Combat Aerial Vehicle (UCAV) formation is expected to bring out strengths in Beyond-Visual-Range (BVR) air combat. Although Multi-Agent Reinforcement Learning (MARL) shows outstanding performance in cooperative decision-making, it is challenging for existing MARL algorithms to quickly converge to an optimal strategy for UCAV formation in BVR air combat where confrontation is complicated and reward is extremely sparse and delayed. Aiming to solve this problem, this paper proposes an Advantage Highlight Multi-Agent Proximal Policy Optimization (AHMAPPO) algorithm. First, at every step, the AHMAPPO records the degree to which the best formation exceeds the average of formations in parallel environments and carries out additional advantage sampling according to it. Then, the sampling result is introduced into the updating process of the actor network to improve its optimization efficiency. Finally, the simulation results reveal that compared with some state-of-the-art MARL algorithms, the AHMAPPO can obtain a more excellent strategy utilizing fewer sample episodes in the UCAV formation BVR air combat simulation environment built in this paper, which can reflect the critical features of BVR air combat. The AHMAPPO can significantly increase the convergence efficiency

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of the strategy for UCAV formation in BVR air combat, with a maximum increase of 81.5% relative to other algorithms.

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1. Introduction

With the development of air combat weapons, the air engagement area is gradually expanded to BVR, and BVR air combat plays a significant role in contemporary air combat. However, it is difficult for pilots to make outstanding decisions cooperatively to strive for the advantages of first view, first shot, and first kill in BVR air combat. UCAV formation can replace manned combat aerial vehicle formation in BVR air combat,^{1,2} the critic of which is the autonomous cooperative decision-making ability of UCAV formation.

At present, cooperative decision-making oriented to air combat has been widely studied,^{3–7} and the methods commonly used can be divided into the game method and intelligent optimization method. The game method mainly includes matrix game, influence diagram game, differential game, and others. Li et al.⁸ develop the dimensionality reduction based matrix game algorithm to work out large-scale matrix games promptly. Nevertheless, considering the air combat scenarios constructed in that paper are relatively simple, it is unknown whether the algorithm has the local optimal problem in the complex BVR air combat with a long confrontation cycle. Influence diagram game⁹ and differential game¹⁰ are mostly utilized for decision-making problems in Within-Visual-Range (WVR) air combat. The influence diagram game method needs to establish air combat models that directly express crucial factors. Nonetheless, due to the long calculation time, it cannot meet the real-time requirement of decision-making. The differential game method relies on an accurate mathematical model and can hardly obtain an optimal strategy because of the complex and changeable situation in BVR air combat and the huge amount of calculation. Therefore, the game method has some limitations in solving the cooperative decision-making problem in BVR air combat.

The intelligent optimization method mainly contains particle swarm optimization algorithm, evolutionary algorithm, and Reinforcement Learning (RL)¹¹ algorithm. An improved discrete particle swarm optimization algorithm¹² is designed to deal with the cooperative occupation problem of multi-Unmanned Aerial Vehicle (multi-UAV) involved in BVR air combat. Li et al.¹³ combine genetic algorithm and neural network to handle the target assignment problem of multi-UAV in BVR air combat. However, to some extent, the above heuristic algorithms rely on expert experience data, which is tough to obtain.

In recent years, the Deep Neural Network has stimulated the growth of the RL algorithm, which has achieved remarkable performance in many decision-making domains, such as Atari,¹⁴ Go,¹⁵ and StarCraft II.¹⁶ Besides, the RL algorithm does not need expert experience data, and can meet the requirement of real-time decision-making, so it is more suitable to deal with the decision-making problem in BVR air combat. Even so, there is more research on maneuver decision-making in WVR air combat,^{17–22} relatively less attention on decision-

making in BVR air combat with a more complicated situation, longer decision-making cycle, and more sparse and delayed reward. Piao et al.²³ improve Proximal Policy Optimization (PPO), Yuan et al.²⁴ design an improved Q-network, and Hu et al.²⁵ propose an improved Deep Q-Network (DQN) to realize the decision-making of 1vs.1 BVR air combat. Hu et al.²⁶ integrate situation assessment with DQN to generate cooperative strategies in BVR air combat. Nevertheless, this strategy introduces rules to assist decision-making. Liu et al.²⁷ adopt the Multi-Agent Proximal Policy Optimization (MAPPO)²⁸ to handle the problem of autonomous decision-making for multiple UCAVs in BVR air combat, which does not involve airborne radar and the movement of missiles. The BVR air combat simulation environments for these studies are not realistic enough. On this account, they barely involve the low strategy convergence efficiency problem caused by sparse and delayed reward, which exists in BVR air combat indeed.

It can be seen that the study of cooperative decision-making in BVR air combat based on MARL still has room for improvement. First, the simulation environments of cooperative BVR air combat studied by the existing research are relatively simplified. To be specific, the partially observable battlefield, communication between UCAVs based on the data link, and mid-course guidance phase of the missile are ignored, and they are the major characteristics of cooperative BVR air combat. Partially observable battlefield means that when the target does not meet the search condition of airborne radar or passive detection condition of Electronic Support Measures (ESM), the UCAV cannot obtain information about the target by itself. Communication between UCAVs based on the data link denotes that the UCAV formation can realize the sharing of the target's information with small errors through the data link between UCAVs that meet the communication condition. Mid-course guidance phase of the missile is taking more and more proportion of the missile's flight with the increasing attack distance, and becomes the critic to affect the missile's hit rate in BVR air combat. Consequently, considering the above characteristics, it follows that the cooperative confrontation strategy in BVR air combat should contain the strategy of acquiring the enemy's information and guiding missiles collaboratively through the coordination of switching radar and maneuvering among UCAVs.

Second, in BVR air combat involving the above characteristics, the problem of low strategy convergence efficiency caused by sparse and delayed reward has not been solved. Specifically, the ultimate goal of BVR air combat is to destroy all of the enemy's UCAVs, but the process is extremely complicated because it involves search, tracking, launch, mid-course guidance, terminal guidance, and others. And there is a strong correlation among all the above phases, which puts forward higher requirements for the timing of decision-making. Furthermore, in the course of strategy optimization utilizing the MARL algorithm, only some quite small rewards can be received before the enemy's UCAV is destroyed, and a large

reward can be obtained at the moment of the destruction. But the result of the destruction is contributed by previous serialization decisions, which makes the strategy convergence extremely difficult and slow, let alone the convergence of cooperative confrontation strategy.

Presently, the main method to deal with the problem of sparse and delayed reward in the field of RL is reward shaping, which can be roughly divided into three research directions. One is to encourage exploration,^{29,30} that is, by setting additional reward to encourage the agent to explore the state never experienced. Nonetheless, the space of the UCAV formation BVR air combat simulation environment is pretty large, and the confrontation phases are complex, which can lead to great state space. Obviously, it is tough to converge to the optimal strategy quickly and stably by encouraging exploration directly. Another is to redistribute the final reward,³¹ meaning to make the actor network normally converge by allocating the final reward to the past state-action pairs according to a certain logic. However, how to allocate the reward to the state-action pairs in BVR air combat with a long confrontation period and complicated confrontation process is also a question, which increases the difficulty of solving the problem. The other is the potential-based reward shaping method, which enriches the reward function by designing the potential function.^{32,33} Unlike the WVR air combat involved in Refs. 32, 33, for BVR air combat, which has more complex confrontation phases, this method is difficult to implement. On the contrary, an unreasonable potential function can affect the convergence effect of the MARL algorithm. Accordingly, the reward shaping approach is not suitable or applicable for improving the convergence efficiency of the actor network in the UCAV formation BVR air combat simulation environment. Both from the perspective of pre-war strategy preparation and wartime strategy iterative optimization, converging in a shorter time to get a better cooperative confrontation strategy is significant.³⁴ There is a need to develop a practicable method based on the current MARL algorithms with strong sample efficiency, such as MAPPO, to solve the problem of low strategy convergence efficiency.

This paper focuses on the basic features of UCAV formation BVR air combat and mainly solves the problem of low convergence efficiency of the actor network in BVR air combat. Concretely speaking, the contributions of this paper are as follows:

- (1) A UCAV formation BVR air combat simulation environment, which contains the partially observable battlefield, communication between UCAVs based on the data link, and mid-course guidance phase of the missile, is constructed. So that the major characteristics of BVR air combat can be reflected, and the problem of low strategy convergence efficiency can be exposed.
- (2) An AHMAPPO algorithm is proposed to enhance the sensitivity of the actor network to the potential optimal strategies in the updating process by highlighting the degree to which the best formation exceeds the average of formations in parallel environments. In this way, the actor network can converge to the optimal strategy more efficiently in the UCAV formation BVR air combat simulation environment. The rule-based complex cooperative confrontation strategy for the blue-side

UCAV formation is developed to support the training of the red-side UCAV formation based on the AHMAPPO algorithm.

- (3) By means of the AHMAPPO algorithm proposed in this paper, the 3vs.3 cooperative confrontation strategy with a high win rate is obtained in the UCAV formation BVR air combat simulation environment. The excellent convergence efficiency of the AHMAPPO algorithm has also been proved by comparing it with some state-of-the-art algorithms. Besides, three typical strategies are taken as examples to show the effect of the 3vs.3 cooperative confrontation strategy.

The rest of this paper is organized as follows. [Section 2](#) describes the UCAV formation BVR air combat simulation environment and models the decision-making process of the UCAV formation BVR air combat. [Section 3](#) details the proposed AHMAPPO algorithm and the confrontation strategy for the blue-side formation. [Section 4](#) shows the simulation results. [Section 5](#) summarizes and explains the obtained strategy. [Section 6](#) concludes this paper.

2. Problem formulation

2.1. UCAV formation BVR air combat simulation environment

2.1.1. Scenario

The UCAV formation BVR air combat scenario, generally speaking, is designed as follows: two sides' UCAV formations engage in BVR air combat with no other operational support UCAV in the space of a cylindrical area. The blue-side formation makes decisions according to the rule-based cooperative confrontation strategy designed in this paper, while the red-side formation decides based on the AHMAPPO algorithm. The combat objective of both sides is to kill all the other UCAVs on the battlefield by launching and guiding long-range air-to-air missiles. Within the specified time, the only surviving side or the side with more surviving UCAVs is regarded as the winner. To encourage the algorithm to converge to a more excellent strategy, the blue side is judged to win when the number of remaining UCAVs on both sides is equal.

As for UCAVs' ability, it is assumed that the two sides' UCAVs are homogeneous. Every UCAV can lock the enemy within the search range through airborne radar, thus obtaining accurate information about the enemy. Meanwhile, ESM can be employed to passively acquire the enemy information with error. Moreover, friends can receive or send data through the data link to realize information exchange. The information received through the data link is also subject to small errors compared to the real. The effective distance of both the ESM and the data link can cover the entire battlefield. After obtaining the target's information, the two sides' UCAVs will automatically launch missiles to attack if they meet the launch conditions that the enemy is located within the maximum and minimum launch range. The UCAV is considered dead if it is shot down by the missile, flies out of the battlefield, or the space between another UCAV and itself is less than a certain collision avoidance range.

Two sides' missiles are also homogeneous here. During the mid-course guidance phase, the missile will track in light of the

target's information sent by the guidance UCAV. If the data link between the missile and the guidance UCAV is interrupted, the tracking will be conducted on the strength of the latest target's information, assuming that the target keeps a uniform and straight motion. Therefore, in order to ensure the missile's hit rate, the UCAVs need to guide the missile as much as possible until the missile enters the terminal guidance phase or the missile's fly time reaches the maximum guidance time. What's more, if the range between the target and missile is less than the explosion radius, the missile will automatically explode, and the target will be shot down. The missile fails if it explodes, flies out of the battlefield, or its speed is less than the minimum speed.

2.1.2. UCAV and missile models

The three-degree-of-freedom model is adopted to describe the motion of the UCAV and missile. As shown in Fig. 1, the body coordinate system $O_b x_b y_b z_b$ with the center of gravity O_b of the UCAV as the origin and the inertial coordinate system $O_g x_g y_g z_g$ with the center of the battlefield O_g as the origin are established.

The kinematic and dynamic models of the UCAV are given by:

$$\begin{cases} \dot{x} = V \cos \theta \cos \psi \\ \dot{y} = V \cos \theta \sin \psi \\ \dot{z} = -V \sin \theta \end{cases} \quad (1)$$

$$\begin{cases} \dot{V} = g(n_x - \sin \theta) \\ \dot{\theta} = \frac{g}{V}(n_z \cos \phi - \cos \theta) \\ \dot{\psi} = \frac{g n_z \sin \phi}{V \cos \theta} \end{cases} \quad (2)$$

where (x, y, z) represents the position of the UCAV, n_x and n_z respectively denote the longitudinal and normal overload of the UCAV, V is the speed of the UCAV, and g is the acceleration of gravity. (θ, ϕ, ψ) represents the attitude angle of the UCAV, wherein the pitch angle θ is the included angle between the axis $O_b x_b$ and the horizontal plane $O_g x_g y_g$, and the angle value is set to be positive when the UCAV's nose is tilted up. The roll angle ϕ is the included angle between the symmetry plane of the UCAV and the plumb plane $O_b x_b z'_g$ containing the axis $O_b x_b$, and the angle value is positive when the UCAV rolls to the right. The yaw angle ψ is the included angle between the projection of the axis $O_b x_b$ on the horizontal plane $O_g x_g y_g$ and the axis $O_g x_g$, and the angle value is positive when the UCAV yaws to the right.

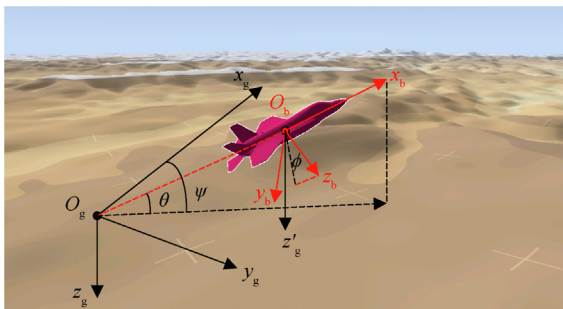


Fig. 1 Three-degree-of-freedom model of UCAV.

The kinematic and dynamic models of the missile are the same as those of the UCAV, which are not detailed here. In the mid-course guidance phase, if the UCAV is located within the maximum guidance angle behind the tail of the missile, the data link between the UCAV and the missile can be established, and the UCAV can send the target's information to guide the missile. The missile utilizes the proportional guidance method to track the target during the flight.

2.1.3. Radar, ESM, and error models

The airborne radar's search range is modeled using the basic radar equation:³⁵

$$\begin{cases} R_{\max} = \left[\frac{P_t G^2 \sigma_{\text{res}}}{(4\pi)^3 P_{\min}} \right]^{1/4} \\ \alpha_{\max} = \beta_{\max} = \frac{70}{180} \pi \end{cases} \quad (3)$$

where P_t is the radar transmitting power, G is the gain of the radar antenna, σ_{res} represents the radar cross-section of the target, $P_{\min}$ is the minimum detectable signal of radar, $R_{\max}$ denotes the maximum range of radar action on the target with the radar cross-section σ_{res} , $\alpha_{\max}$ and $\beta_{\max}$ represent the maximum search angle in yaw and pitch direction of the radar respectively.

The passive detection range equation of the ESM is similar to the basic radar equation. The difference is that the one-way signal radiated by the target's radar can be detected passively by the ESM when the UCAV is located within the maximum search angle of the target's radar.³⁵ Compared with radar search, the advantage is that the detection range of the ESM is wider, and the disadvantage is the low precision, where the detection distance error is proportional to the true distance, and both the detection distance and azimuth error are inversely proportional to the number of UCAVs that can detect the target passively by the ESM in this paper. On that account, in order to reflect the characteristics of the ESM's detection error, the passive detection distance and azimuth error are set as:

$$\begin{cases} \Delta R = \pm \frac{D_{\text{error esm}}}{n_{\text{esm}}} D_{\min} \\ \Delta \alpha = \Delta \beta = \pm \frac{A_{\text{error esm}}}{n_{\text{esm}}} \end{cases} \quad (4)$$

where ΔR is the detection distance error of the ESM, $\Delta \alpha$ and $\Delta \beta$ represent the detection azimuth error of the ESM in yaw and pitch direction, respectively, $D_{\text{error esm}}$ is the coefficient of the detection distance error, $A_{\text{error esm}}$ is the maximum detection azimuth error, n_{esm} indicates the number of UCAVs that can detect the target passively by the ESM, and $D_{\min}$ represents the minimum distance between the target and UCAVs which involve in the collaborative ESM detection.

Furthermore, friends in formation can also communicate with each other through the data link to share the information searched or detected by themselves. Nevertheless, considering the delay of information transmission, the coordinate conversion between UCAVs, and other factors, there are a few errors in the information shared by UCAVs through the data link. It is assumed that errors are proportional to the distance between UCAVs sharing information:

$$\begin{cases} \Delta P = \pm \frac{R_{\text{link}}}{R_{\text{link max}}} \cdot D_{\text{error link}} \\ \Delta \Phi = \pm \frac{R_{\text{link}}}{R_{\text{link max}}} \cdot A_{\text{error link}} \end{cases} \quad (5)$$

where ΔP and $\Delta\Phi$ are the error of shared position and attitude information respectively, R_{link} represents the distance between UCAVs sharing information, $R_{\text{link}_{\text{max}}}$ denotes the maximum communication distance between UCAVs, $D_{\text{error}_{\text{link}}}$ and $A_{\text{error}_{\text{link}}}$ are the maximum error of the shared position and attitude information, respectively.

2.1.4. Guidance mode

In view of the characteristics and means of cooperative confrontation in BVR air combat, the following four cooperative guidance modes are summarized by taking the cooperation between two red-side UCAVs as an example:

(1) One UCAV launches and guides, the other searches

One UCAV receives the target's information from the other UCAV, which is responsible for searching and tracking the target, and launches a missile if the condition is met. In the following time, the UCAV that launches the missile will provide it with mid-course guidance, as depicted in Fig. 2 (The red dotted line represents that the UCAV is providing guidance for the missile). This cooperative guidance mode expands the threat range of the UCAV that launches the missile when only the other UCAV can search the target.

(2) One UCAV launches, the other searches and guides

One UCAV obtains the target's information provided by the other UCAV, which is in charge of searching and tracking the target, and launches a missile if the condition is met. Afterward, the UCAV that searches and tracks the target will provide the missile with mid-course guidance, as shown in Fig. 3. This cooperative guidance mode enables the UCAV to choose to leave the battlefield, threaten other targets or remain silent after launching the missile, which enhances the possibility and flexibility of tactical changes and improves the adaptability of the UCAV formation to the battlefield.

(3) Alternate guidance

This mode represents the switch between the above two guidance modes. Specifically, one UCAV launches and guides the missile and the other searches for the target at first. When one UCAV receives other orders or encounters threats, it has to abandon the mid-course guidance. At this moment, the

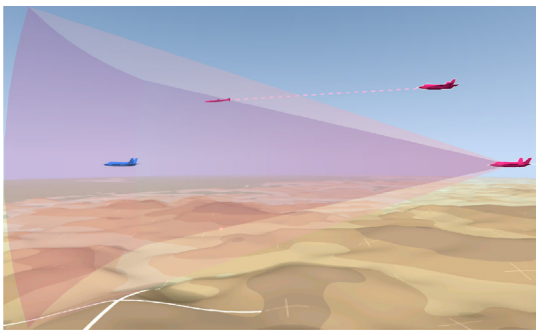


Fig. 2 Mode of “One UCAV launches and guides, the other searches”.

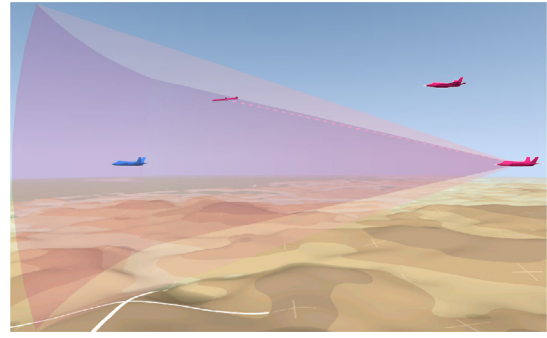


Fig. 3 Mode of “One UCAV launches, the other searches and guides”.

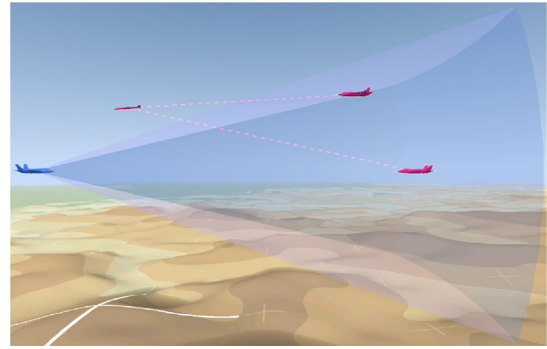


Fig. 4 Mode of “Collaborative guidance by ESM”.

other UCAV that can communicate with the missile takes over to provide guidance to the missile. By carrying out guidance handover irregularly, it is tough for the enemy to continuously lock UCAVs in formation. This cooperative guidance mode can not only ensure guidance accuracy but also improve the survival rate of the formation.

(4) Collaborative guidance by ESM

The UCAVs in formation do not turn on radar and detect the target passively by ESM, which is the only difference between this mode and the above three modes, as shown in Fig. 4. In the process of this mode, the UCAVs that can detect the target passively by ESM in formation keep silent, which is safer than the other guidance modes. However, the errors in the target's information will affect the guidance accuracy, which will reduce the missile's hit rate. Nevertheless, the formation can make use of multi-UCAV ESM for collaborative passive detection, which can make up for the lack of information accuracy to a certain degree. It should be pointed out that, as seen in Fig. 4, two red dotted lines are adopted to indicate that there are two UCAVs in the mode of “Collaborative guidance by ESM” and satisfy the conditions of ESM detection and guidance, but only one UCAV can guide the missile using the collaborative passive detection results actually.

2.2. Decentralized partially observable Markov decision process

In BVR air combat, the battlefield information is partially observable. Each UCAV can only make decisions on the basis

of the information acquired by itself (including the shared information from the data link, the error of which is related to the location of the UCAV in communication and, therefore, varies). As a consequence, the decentralized partially observable Markov decision process (Dec-POMDP) is applied to model the cooperative decision-making problem for UCAV formation in BVR air combat.

The Dec-POMDP can be expressed as tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{R}, \mathcal{P}, n, \gamma \rangle$. Where $\mathcal{A}$ is the action space shared by n agents, $o^i = \mathcal{O}(s; i)$ represents the local observation of the global state $s \in \mathcal{S}$ observed by the agent i , $\mathcal{R}(s, (a_1, \dots, a_n))$ denotes the shared reward function, $\mathcal{P}(s'|s, (a_1, \dots, a_n))$ indicates the transition probability from s to s' given the joint action $(a_1, \dots, a_n)$, and γ is the discount factor.

2.2.1. Action space

The action space of the agent is set as $\mathcal{A} = [\psi_{\text{desire}}, \theta_{\text{desire}}, c_r]$, where ψ_{desire} is the expected yaw angle and θ_{desire} is the expected pitch angle. $c_r \in \{0, 1\}$ represents radar status choice, where 0 is off and 1 is on. ψ_{desire} and θ_{desire} here are discretized and combined into a series of expected angle pairs $\{\psi_{\text{desire}}, \theta_{\text{desire}}\}$, as shown in Table 1.

According to the arrangement of the expected angle pairs in Table 1, the actor network of the agent only needs to output the index number. Similarly, for radar status selection, it only needs to output 0 or 1 indicating radar off or radar on, respectively.

2.2.2. Global state

The global state of agent i that can reflect the characteristics of the simulation environment at a certain moment contains not only agent i 's own state information but also the relative state information and is designed as follows:

$$\mathbf{S}^i = \begin{bmatrix} (x^i, y^i, z^i, V_x^i, V_y^i, V_z^i, \theta^i, \phi^i, \psi^i, \text{RS}^i, \text{ES}^i, \text{RF}^i, \text{EF}^i, \text{BL}^i, \text{BT}^i, \text{RO}^i) \\ (\Delta x^{ji}, \Delta y^{ji}, \Delta z^{ji}, \Delta V_x^{ji}, \Delta V_y^{ji}, \Delta V_z^{ji}, \text{RS}^j, \text{ES}^j, \text{RF}^j, \text{EF}^j, \text{BL}^j, \text{BT}^j, \text{RO}^j) |_{j \in i-} \\ (\Delta x^{ki}, \Delta y^{ki}, \Delta z^{ki}, \Delta V_x^{ki}, \Delta V_y^{ki}, \Delta V_z^{ki}, \text{RO}^k, \text{RG}^k, \text{EG}^k, \text{GS}^k, \text{FR}^k, \alpha^{ki}, \beta^{ki}, \text{BG}^k) |_{k=1, \dots, n_{\text{RM}}} \\ (\Delta x^{pi}, \Delta y^{pi}, \Delta z^{pi}, \Delta V_x^{pi}, \Delta V_y^{pi}, \Delta V_z^{pi}, \text{RS}^p, \text{ES}^p, \text{RF}^p, \text{EF}^p, \text{BL}^p, \text{BT}^p, \text{RO}^p, \alpha^{pi}, \beta^{pi}, \text{ST}^p) |_{p=1, \dots, n_{\text{B}}} \\ (\Delta x^{qi}, \Delta y^{qi}, \Delta z^{qi}, \Delta V_x^{qi}, \Delta V_y^{qi}, \Delta V_z^{qi}, \text{RO}^q, \text{RG}^q, \text{EG}^q, \text{GS}^q, \text{FR}^q, \alpha^{qi}, \beta^{qi}, \text{RL}^q) |_{q=1, \dots, n_{\text{BM}}} \end{bmatrix} \quad (6)$$

where RS^i indicates the number of enemy UCAVs which can search agent i by the radar, ES^i represents the number of enemy UCAVs which can detect agent i by the ESM, RF^i denotes the number of missiles launched by the agent i at targets searched by its own or friends' radar, EF^i represents the number of missiles launched by the agent i at targets detected by its own or friends' ESM, and BL^i is the total time that the

agent i has been locked by enemy's missiles. BT^i indicates whether there are missiles targeting the agent i on the current battlefield, and if so $\text{BT}^i = 1$, otherwise $\text{BT}^i = 0$. RO^i denotes whether the agent i turns on the radar, and if so $\text{RO}^i = 1$, otherwise $\text{RO}^i = 0$. $\Delta x^{ab}, \Delta y^{ab}, \Delta z^{ab}, \Delta V_x^{ab}, \Delta V_y^{ab}$, and ΔV_z^{ab} represent the relative position and velocity of each axis in the inertial coordinate system between a and b . α^{ab} and β^{ab} denote the angles of a with respect to b in the yaw and pitch directions, respectively. $i-$ indicates agents in the formation other than the agent i . RG^k represents the total time that the missile k has been guided by the guidance agent which can search the target by its own or friends' radar. EG^k represents the total time that the missile k has been guided by the guidance agent which can passively detect the target by its own or friends' ESM. GS^k denotes the guidance situation of the missile k by each agent on the current battlefield, which uses the One-Hot Encoding method to indicate that whether the agent in formation is guiding the missile at the moment. If so, the corresponding element is set to 1, otherwise set to 0. FR^k represents the launch distance of the missile k , that is, the relative distance between the agent and the target when the missile is launched. n_{RM} and n_{BM} are the maximum number of missiles launched by the red-side and blue-side UCAVs respectively, and not failed in the environment. BG^k indicates whether the missile k is guided by any agents, and if so $\text{BG}^k = 1$, otherwise $\text{BG}^k = 0$. $\text{ST}^p \in \{0.4, 0.6, 0.8, 1.0\}$ represents the reliability of the data acquisition path of the UCAV p . Its value corresponds to the four ways of passive detection by friends' ESM, passive detection by own ESM, search by friends' radar, and search by own radar. n_{B} is the initial number of UCAVs in the blue-side formation. RL^q denotes whether the agent i is in the effective range of the missile q 's radar. It should be pointed out that there will be dead agents

or no missile that has been launched and not failed in the environment. At this time, all the corresponding state information will be set to 0. Otherwise, the above state space without error is given as the global state information shared by all agents in the environment.

2.2.3. Local observation

The local observation of the agent i designed in this paper is:

$$\mathbf{o}^i = \begin{bmatrix} (x^i, y^i, z^i, V_x^i, V_y^i, V_z^i, \theta^i, \phi^i, \psi^i, \mathbf{RS}^i, \mathbf{RF}^i, \mathbf{EF}^i, \mathbf{BL}^i, \mathbf{RO}^i) \\ (\Delta x^{ji}, \Delta y^{ji}, \Delta z^{ji}, \Delta V_x^{ji}, \Delta V_y^{ji}, \Delta V_z^{ji}, \mathbf{RS}^j, \mathbf{RF}^j, \mathbf{EF}^j, \mathbf{BL}^j, \mathbf{RO}^j) |_{j \in i-} \\ (\Delta x^{ki}, \Delta y^{ki}, \Delta z^{ki}, \Delta V_x^{ki}, \Delta V_y^{ki}, \Delta V_z^{ki}, \mathbf{RO}^k, \mathbf{RG}^k, \mathbf{EG}^k, \mathbf{GS}^k, \mathbf{FR}^k, \alpha^{ki}, \beta^{ki}) |_{k=1, \dots, n_{\text{RM}}} \\ (\Delta x^{pi}, \Delta y^{pi}, \Delta z^{pi}, \Delta V_x^{pi}, \Delta V_y^{pi}, \Delta V_z^{pi}, \mathbf{RS}^p, \mathbf{ES}^p, \mathbf{BT}^p, \alpha^{pi}, \beta^{pi}, \mathbf{ST}^p, \mathbf{RS}^{pi}) |_{p=1, \dots, n_{\text{B}}} \\ (\Delta x^{qi}, \Delta y^{qi}, \Delta z^{qi}, \Delta V_x^{qi}, \Delta V_y^{qi}, \Delta V_z^{qi}, \alpha^{qi}, \beta^{qi}) |_{q=1, \dots, n_{\text{BM}}} \end{bmatrix} \quad (7)$$

where $\mathbf{RS}^{pi}$ denotes whether the enemy's UCAV p can search agent i through radar. Obviously, compared with $\mathcal{S}^i$, there is some information about red-side and blue-side UCAVs that the agent $r_{i,\max}$ can not observe due to the partially observable battlefield characteristic in the UCAV formation BVR air combat simulation environment. When the agent $r_{i,\min}$ can obtain the information of a blue side UCAV through the radar, ESM, or data link, the corresponding variable in the \(* MERGEFORMAT will be assigned the corresponding value, otherwise, it will remain 0.

2.2.4. Reward

It should be noted that the meaning of the reward is pointing out the final goal to the agent rather than directly telling the agent how to achieve it. Therefore, the reward should be designed according to the objective events that can reflect the current situation on the battlefield. In this paper, the reward is designed as shown in Table 2.

The formation reward at each step can be calculated by adding the items in Table 2 in light of the events that are happening. Considering that the detection accuracy of the ESM is related to the number of the UCAVs participating in the passive detection when calculating the reward related to the ESM, the corresponding item in Table 2 will be multiplied by the number of the UCAVs that can passively detect the target through the ESM at that moment.

3. Decision-making method

3.1. MAPPO

As a popular MARL algorithm, the MAPPO, developed from the PPO³⁶ algorithm, adopts Centralized Training and Decentralized Execution (CTDE) architecture and has been

approved can achieve better or comparable sample complexity as well as substantially faster running time compared to off-policy baselines in the majority of environments. The MAPPO's actor network π_{ω} is optimized by maximizing the loss function:

Table 2 Reward settings.

Event	Reward	Event	Reward
A blue-side UCAV is shot down	10.0	A red-side UCAV is shot down	-10.0
A blue-side UCAV flies out of battlefield	10.0	A red-side UCAV flies out of battlefield	-10.0
A red-side UCAV launches a missile at a target searched by own or friend's radar	1.0	A blue-side UCAV launches a missile at a target searched by own or friend's radar	-1.0
A red-side UCAV launches a missile at a target detected by own or friend's ESM	0.5/3	A blue-side UCAV launches a missile at a target detected by own or friend's ESM	-0.5/3
A red-side UCAV searches a target by own radar	0.001	A blue-side UCAV searches a target by own radar	-0.001
A red-side UCAV detects a target by own ESM	0.0005/3	A blue-side UCAV detects a target by own ESM	-0.0005/3
A red-side UCAV guides a missile	1/600	A blue-side UCAV guides a missile	-1/600
A blue-side UCAV is locked by a red-side missile	0.003	A red-side UCAV is locked by a blue-side missile	-0.003

Table 1 Expected angle pairs and index number.

Index number	Angle pair(rad)	Index number	Angle pair(rad)	Index number	Angle pair(rad)
0	$\{-3\pi/4, -\pi/6\}$	6	$\{-3\pi/4, 0\}$	12	$\{-3\pi/4, \pi/6\}$
1	$\{-\pi/4, -\pi/6\}$	7	$\{-\pi/4, 0\}$	13	$\{-\pi/4, \pi/6\}$
2	$\{0, -\pi/6\}$	8	$\{0, 0\}$	14	$\{0, \pi/6\}$
3	$\{\pi/4, -\pi/6\}$	9	$\{\pi/4, 0\}$	15	$\{\pi/4, \pi/6\}$
4	$\{3\pi/4, -\pi/6\}$	10	$\{3\pi/4, 0\}$	16	$\{3\pi/4, \pi/6\}$
5	$\{\pi, -\pi/6\}$	11	$\{\pi, 0\}$	17	$\{\pi, \pi/6\}$

$$L(\omega) = \left[\frac{1}{Bn} \sum_{j=1}^B \sum_{i=1}^n \min \left(r_{\omega,j}^i A_j^i, \text{clip} \left(r_{\omega,j}^i, 1 - \kappa, 1 + \kappa \right) A_j^i \right) \right] + \ell \frac{1}{Bn} \sum_{j=1}^B \sum_{i=1}^n S \left[\pi_{\omega} \left(o_j^i \right) \right] \quad (8)$$

where B represents the batch size, $r_{\omega,j}^i = \pi_{\omega} \left(a_j^i | o_j^i \right) / \pi_{\omega_{\text{old}}} \left(a_j^i | o_j^i \right)$, A_j^i is computed using the Generalized Advantage Estimation (GAE)³⁷ method, κ is the clip parameter, S represents the entropy of the strategy, $A^* = \alpha * \text{clip} \left(N \left(\frac{R_{t,\text{max}} - R_{t,\text{ave}}}{(1 - 1/n_{\text{env}})(R_{\text{max}} - R_{\text{min}})}, -1, 1 \right), -1, 1 \right)$ is the local observation of agent $R_{t,\text{max}}$ at step $R_{t,\text{min}}$ in batch, and ℓ is the entropy coefficient hyperparameter. In addition, the optimization goal of the critic network V_{ω} is to minimize the loss function:

$$L(\varpi) = \frac{1}{Bn} \sum_{j=1}^B \sum_{i=1}^n \left(\max \left[\left(V_{\varpi} \left(s_j^i \right) - \hat{R}_j \right)^2, \left(\text{clip} \left(V_{\varpi} \left(s_j^i \right), V_{\varpi_{\text{old}}} \left(s_j^i \right) - \zeta, V_{\varpi_{\text{old}}} \left(s_j^i \right) + \zeta \right) - \hat{R}_j \right)^2 \right] \right) \quad (9)$$

where ζ is the clip parameter, s_j^i is the global state of agent n_{env} at step A^* in batch, and $\hat{R}_j$ denotes the discounted reward-to-go.

3.2. AHMAPPO

3.2.1. Advantage highlight

Different from other game scenarios, for the reason that it takes to search, tracking, launch, mid-course guidance, and other phases for the UCAV to hit the target and obtain the final large reward, the reward in BVR air combat is extremely sparse and delayed. What's more, there are close relationships between all the above phases in the confrontation process, so it is difficult to converge to an actor network efficiently that can make the optimal serialization decisions in the BVR air combat with drastic situation changes. Consequently, to quickly find the optimization direction of the optimal strategy under sparse and delayed rewards, it is quite necessary to pay as

much attention as possible to the outstanding decisions displayed by each agent in the training process.

Nevertheless, in the training process where multiple agents are homogeneous, the existing algorithm simply optimizes the strategy by averaging the performance of each agent, which ignores the outstanding performance of the individual. In other words, the poor performance of some agents reduces the overall reward, which tends to make the optimization of the actor network excessively deviate from the right direction, delay the convergence speed, or even fail to converge to the optimal strategy. Of course, it should be noted that the above situation does not exclude the possibility that the current strategy is locally optimal. But the current strategy is effective in this case, after all. Simply selecting the average performance of agents for optimization is easy to make the actor network hard to find the correct optimization direction and collapse, especially in the early training process. This is particularly acute in long decision-making cycle confrontation environments with sparse and delayed reward, such as BVR air combat.

As a typical MARL algorithm, the MAPPO has the above problem. As can be seen in Eq. (8), the loss function of the actor network adds and averages the advantage of each agent in the batch during optimization. This will easily lead to the high advantage of excellent agents being pulled down, in this way, affecting the optimization direction of the actor network. For example, suppose that the rewards received by three agents at a certain step are respectively r_1, r_2 and r_3 . The average of the above three agents' rewards is r_{ave} , and $r_1 > r_2 > r_{\text{ave}} > r_3$, as shown in Fig. 5(a). At the same time, the advantages of three agents at this step are A_1, A_2 and A_3 respectively. Since the MAPPO adopts the CTDE architecture for strategy optimization, the value of agents' critic network will be the same normally. Then, the average of the above three agents' advantages is A_{ave} , and the advantages satisfy $A_1 > A_2 > A_{\text{ave}} > A_3$, as shown in Fig. 5(b). Obviously, if the loss function is calculated only by adding and averaging the advantages of all agents, due to the existence of Agent 3 with the worst performance in formation, it will cause the high advantage of Agent 1 to be neutralized to a large extent. However, it is evident that Agent 1 performs well at this step, and its performance is far above the average level of all agents. This information should not be ignored, especially in the long

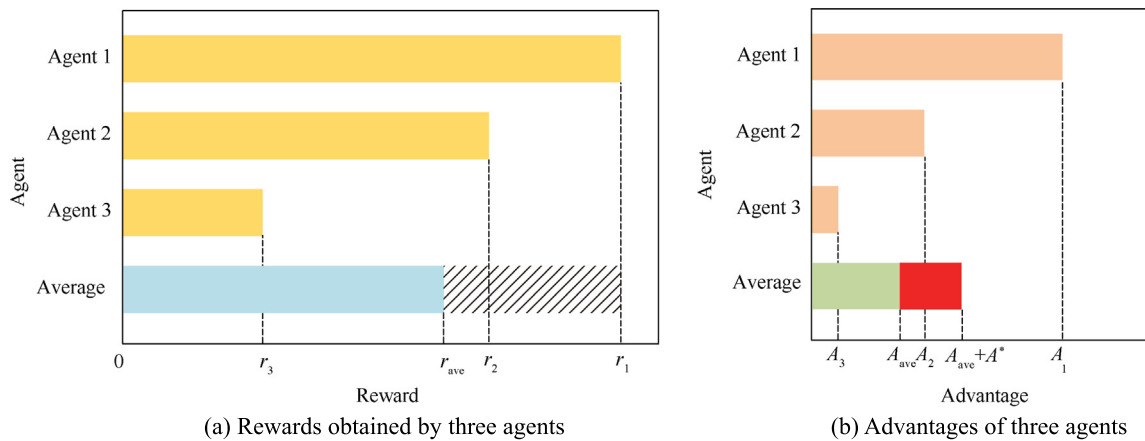


Fig. 5 Impact of poor performance of some agents.

decision-making cycle scenario where the reward is sparse and delayed. Accordingly, to more comprehensively reflect the performance of the current actor network, a compensation item is proposed in this paper, corresponding to the part of the red dotted box in Fig. 5(b):

$$A^* = \alpha * \text{clip} \left(N \left(\frac{r_1 - r_{\text{ave}}}{(1 - 1/3)(r_{\text{max}} - r_{\text{min}})}, \sigma \right) \right. \\ \left. \middle/ \left(\frac{r_1 - r_{\text{ave}}}{(1 - 1/3)(r_{\text{max}} - r_{\text{min}})} + 3\sigma \right), -1, 1 \right) \quad (10)$$

where r_{max} and r_{min} respectively represent the maximum and minimum rewards that the agent can receive, σ is the variance of rewards obtained by all agents in formation at the current step, N denotes the Normal Distribution sampling, and α represents a regulatory factor used to match the value of the original advantage. Specifically, this method calculates the degree to which the agent with the best performance in the formation exceeds the average level $r_1 - r_{\text{ave}}$ and divides it by its possible maximum value as the mean of the Normal Distribution. And the variance σ is taken as the variance of the Normal Distribution. Meanwhile, the randomness of the Normal Distribution sampling can prevent the strategy from falling into local optimality to some extent. In addition, to prevent that there's a quite large or small value to this term, the result of the Normal Distribution sampling is shrunk and clipped to the range $(-1, 1)$. The general formula of A^* can be summarized as follows:

$$A^* = \alpha * \text{clip} \left(N \left(\frac{r_{t,\text{max}} - r_{t,\text{ave}}}{(1 - 1/n)(r_{\text{max}} - r_{\text{min}})}, \sigma_t \right) \right. \\ \left. \middle/ \left(\frac{r_{t,\text{max}} - r_{t,\text{ave}}}{(1 - 1/n)(r_{\text{max}} - r_{\text{min}})} + 3\sigma_t \right), -1, 1 \right) \quad (11)$$

where $r_{t,\text{max}}$ and $r_{t,\text{min}}$ respectively represent the maximum and minimum rewards obtained by agents in the formation at step t , $r_{t,\text{ave}}$ and σ_t denote the average and variance of rewards obtained by all agents in formation at step t respectively, and n indicates the number of agents in this formation.

The above calculation needs the environment to provide individual reward for each agent. Unfortunately, the formation reward is tough to accurately allocate among agents in many environments, for example, the UCAV formation BVR air combat simulation environment. To be specific, one UCAV locks the target and launches a missile based on the target's

information received by itself, and the missile finally hits the target after another UCAV or even multiple UCAVs participate in the mid-course guidance process at different times. The reward for destroying the target is hard to be divided among the UCAVs involved. Therefore, like many other multi-agent game environments, the UCAV formation BVR air combat simulation environment designed in this paper only provides the reward received by the UCAV formation. Given this, this paper also puts forward a calculation method of A^* for the case of formation reward only:

$$A^* = \alpha \text{clip} \left(N \left(\frac{R_{t,\text{max}} - R_{t,\text{ave}}}{(1 - 1/n_{\text{env}})(R_{\text{max}} - R_{\text{min}})}, \sigma_t^R \right) \right. \\ \left. \middle/ \left(\frac{R_{t,\text{max}} - R_{t,\text{ave}}}{(1 - 1/n_{\text{env}})(R_{\text{max}} - R_{\text{min}})} + 3\sigma_t^R \right), -1, 1 \right) \quad (12)$$

where $R_{t,\text{max}}$ and $R_{t,\text{min}}$ represent the maximum and minimum rewards received by formations in parallel environments at step t respectively, $R_{t,\text{ave}}$ and σ_t^R denote the average and variance of rewards obtained by formations in parallel environments at step t respectively, R_{max} and R_{min} indicate the maximum and minimum rewards that the formation can receive respectively, and n_{env} denotes the number of parallel environments. Concretely speaking, similar to the idea of Eq. (11), the formation rewards in different parallel environments with different random seeds in the process of strategy optimization are selected for comparison, and the same method is used to highlight the extent to which the best formation in different parallel environments outperforms the average. On this account, it can prevent the outstanding performance of the current strategy from being submerged and highlight its advantages.

By means of introducing A^* into the process of actor network optimization in the original MAPPO, the AHMAPPO algorithm is proposed in this paper. Its architecture is shown in Fig. 6.

where $h_{t,\pi}$ and $h_{t,V}$ represent the hidden states of the actor and critic network respectively. The green arrows and components are only required during centralized training. The red arrows and components are the innovations of the AHMAPPO and are also utilized during centralized training.

3.2.2. Policy iteration

Both the actor and critic network are composed of Multilayer Perceptron (MLP) and Recurrent Neural Network (RNN), as

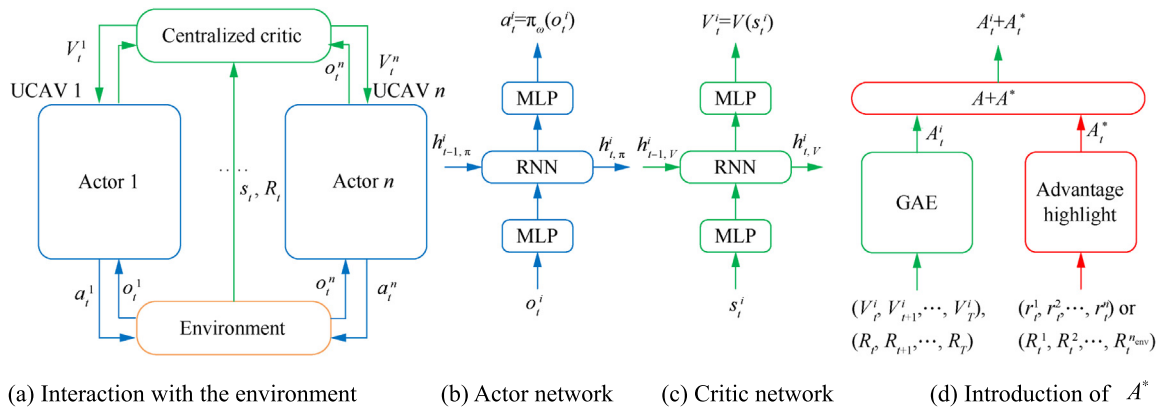


Fig. 6 Architecture of AHMAPPO.

shown in Fig. 6. This paper assumes that both sides' UCAVs are homogeneous, so the agents' actor and critic networks parameters are shared here respectively to speed up the training. In the training process, the update logic of the critic network is consistent with the original MAPPO, but the loss function of the actor network is different as follows:

$$L(\omega) = \left[\frac{1}{Bn} \sum_{j=1}^B \sum_{i=1}^n \min \left(r_{\omega,j}^i (A_j^i + A_j^*), \right. \right. \\ \left. \left. \text{clip} \left(r_{\omega,j}^i, 1 - \kappa, 1 + \kappa \right) (A_j^i + A_j^*) \right) \right] \\ + \ell \frac{1}{Bn} \sum_{j=1}^B \sum_{i=1}^n S[\pi_{\omega}(o_j^i)] \quad (13)$$

As shown in Eq. (13), owing to the fact that A_j^* is independent of the actor network parameter ω , the mathematical expectation of the gradient of Eq. (13) and Eq. (8) are equal, which is similar to introducing baseline in the actor network updating process of the MARL algorithm and has been proven in Ref. 38. As a consequence, A_j^* will not affect the convergence ability of the original MAPPO, on the contrary, as an additional compensation item, it can make the algorithm pay attention to the advantages of the current strategy and converge faster in the process of optimizing the actor network. The pseudo-code of the AHMAPPO is shown in Algorithm 1.

Algorithm 1. AHMAPPO

```

Initialize  $\omega$  and  $\varpi$ , the parameters for actor and critic network
respectively, using Orthogonal initialization
1. for episode = 1 :  $M$  do
2.   Set data buffer  $D = \{\}$  and trajectory  $\tau = []$ 
3.   for  $t = 1 : T$  do
4.     Select action  $a_t^i \sim \pi_{\omega}(o_t^i)$  for each agent  $i$  and get hidden
state  $h_{t,\pi}^i$ 
5.     Compute critic value  $V_t^i = V_{\varpi}(s_t^i)$  for each agent  $i$  and
get hidden state  $h_{t,V}^i$ 
6.     if the environment can provide  $r_t^i$  for each agent  $i$  then
7.       Execute actions, receive  $r_t^i, s_{t+1}^i, o_{t+1}^i$  for each agent  $i$ 
8.       Compute  $A_t^i$  using the advantage highlight method
based on  $r_t^i|_{i=1,\dots,n}$ 
9.       Add  $\left[ \left( s_t^i, o_t^i, h_{t,\pi}^i, h_{t,V}^i, a_t^i, s_{t+1}^i, o_{t+1}^i, r_t^i \right) |_{i=1,\dots,n} \right]$  to  $\tau$ 
10.    end if
11.    if the environment can only provide  $R_t$  for formation then
12.      Execute actions, receive  $R_t^i, s_{t+1}^i, o_{t+1}^i$  for each agent  $i$  in
parallel environment  $j$ 
13.      Compute  $A_t^i$  using the advantage highlight method
based on  $R_t^i|_{j=1,\dots,n_{env}}$ 
14.      Add  $\left[ \left( s_t^i, o_t^i, h_{t,\pi}^i, h_{t,V}^i, a_t^i, s_{t+1}^i, o_{t+1}^i \right) |_{i=1,\dots,n}, R_t^i |_{j=1,\dots,n_{env}} \right]$ 
to  $\tau$ 
15.    end if
16.  end for
17. Compute original advantage estimate using GAE on  $\tau$ 
18. Split  $\tau$  into chunks and add to  $D$ 
19. Sample mini-batch  $d$  from  $D$ 
20. Adam update  $\omega$  on  $L(\omega)$  with  $d$ 
21. Adam update  $\varpi$  on  $L(\varpi)$  with  $d$ 
22. end for

```

3.3. Strategy for blue-side formation

In order to better realize the training of agents in the UCAV formation BVR air combat simulation environment, a complex strategy for the blue-side formation is designed here, which is on the strength of rules and remains unchanged during simulation. Specifically, it includes the radar strategy and maneuver strategy, as shown in Fig. 7.

3.3.1. Radar strategy

The radar strategy for the blue-side formation contains four sub-strategies: conventional radar on strategy, radar off strategy, passive detection to radar strategy, and guidance supplementary strategy. The specifics are as follows:

- (1) Conventional radar on strategy means that when there is no threat, passive detection to radar or guidance requirements, a UCAV in the blue-side formation will be selected to turn on the radar to search the battlefield in light of the UCAV identity number (the lowest value first). The conventional radar on strategy not only ensures that the blue-side formation has a certain control over the battlefield but also does not expose the positions of all UCAVs in the formation. Moreover, since the positions of all UCAVs are randomly initialized in different confrontation rounds, there is no lack of randomness in the choice of which UCAV to turn on the radar under different situations.
- (2) Radar off strategy means that when a UCAV receives the RWR warning, it will immediately turn off its radar. The radar off strategy enables the UCAV in the blue-side formation to be silent immediately after receiving a missile threat, maximizing the survival rate of the UCAV.
- (3) Passive detection to radar strategy denotes that when there is a target only passively detected by ESM, on the basis of the target's information detected by ESM, a UCAV in the blue-side formation that can search the target if its radar is on and is closest to the target will be chosen to turn on the radar to obtain the accurate target's information. The passive detection to radar strategy enables the blue-side formation to acquire accurate information about targets as much as possible while avoiding too much exposure of UCAVs' positions.
- (4) Guidance supplementary strategy indicates that when there is a blue-side missile needs mid-course guidance and no UCAV in the blue-side formation can provide the target's information to the missile, a UCAV in the blue-side formation that can search the target if its radar is on and is closest to the target will be selected to turn on the radar. The guidance supplementary strategy makes the blue-side formation provide the missile with as accurate guidance information as possible and improves the hit rate of the missile without exposing too many UCAVs' positions.

Additionally, among the above four sub-strategies, the priority of the radar off strategy is the highest and others are

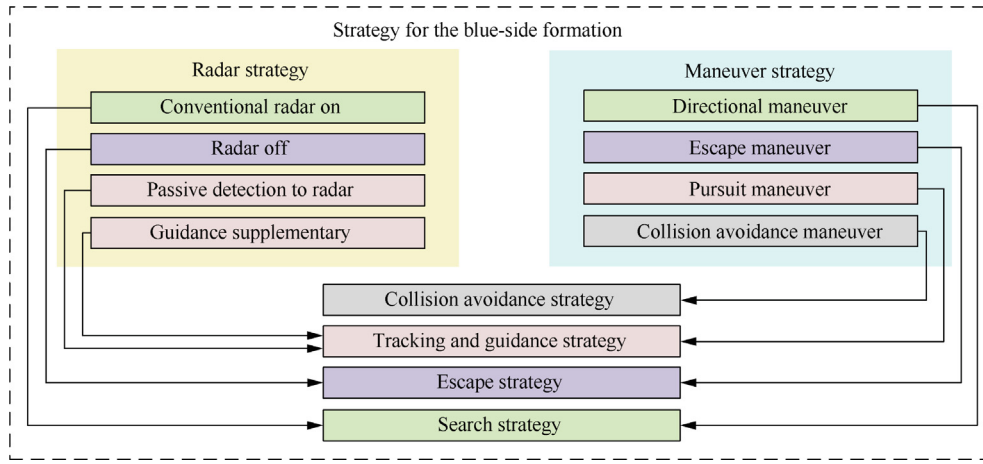


Fig. 7 Strategy for blue-side formation.

employed as needed in different situations. In other words, if a UCAV in blue-side formation is receiving the RWR warning, it will not participate in the execution of other sub-strategies.

3.3.2. Maneuver strategy

The maneuver strategy for the blue-side formation includes four sub-strategies: directional maneuver strategy, escape maneuver strategy, pursuit maneuver strategy, and collision avoidance maneuver strategy. The details are as follows:

- (1) Directional maneuver strategy means that if there is no blue-side UCAV that can search or detect the red-side formation, the expected position of all UCAVs in the blue-side formation will be the center of the battlefield. The directional maneuver strategy enables the blue-side formation to quickly capture the center of the battlefield in the absence of the red-side formation's information.
- (2) Escape maneuver strategy indicates that if a UCAV in the blue-side formation is receiving the RWR warning, it will maneuver to the opposite direction of the red-side missile's signal source to get away from the lock and lower its altitude right away to deplete the missile's energy by the high air density at low altitude. The escape maneuver strategy enables the UCAV, which is in a missile threat, to avoid danger as much as possible.
- (3) Pursuit maneuver strategy denotes that if the information of the red-side formation obtained by the blue-side UCAVs is not empty, each UCAV in the blue-side formation will move towards the red-side UCAV closest to themselves; in other words, the position of the red-side UCAV closest to themselves will be selected as their desired position. The pursuit maneuver strategy makes each UCAV in the blue-side formation efficiently attack the red-side UCAV known to be closest to them.
- (4) Collision avoidance maneuver strategy means that if the distance between two UCAVs in the blue-side formation or between the blue-side UCAV and the red-side UCAV is less than the collision avoidance range, the blue-side UCAV immediately maneuvers to the opposite of their approach direction. The collision avoidance maneuver strategy ensures the UCAV in the blue-side formation from unnecessary sacrifice.

What's more, among the above four sub-strategies, the collision avoidance maneuver strategy has the highest priority, followed by the escape maneuver strategy, pursuit maneuver strategy, and directional maneuver strategy.

The blue-side formation will execute the above radar and maneuver strategies in light of the current situation during simulation. Through the combination of sub-strategies, the collision avoidance strategy, tracking and guidance strategy, escape strategy, and search strategy can be further generated, as shown in Fig. 7.

4. Simulation results

In this part, the AHMAPPO is utilized to train the agents in the UCAV formation BVR air combat simulation environment and compared with other state-of-the-art algorithms, including the MAPPO, Independent PPO (IPPO),³⁹ and QMIX,⁴⁰ which have been shown to have strong performance in strategy convergence and sample efficiency. The MAPPO is also adopted in Ref. 27 to achieve autonomous decision-making for multiple UCAVs in BVR air combat.

Table 3 Environment parameters settings.

Parameter	Value
Number of UCAVs on each side	3
Maximum radar search range (km)	70
Maximum radar search angle (rad)	$\pm 70\pi/180$
Coefficient of the detection distance error	0.12
Maximum detection azimuth error (rad)	$\pm 12\pi/180$
Maximum shared position information error (km)	0.05
Maximum shared attitude information error (rad)	$\pm 5\pi/180$
Number of missiles carried by each UCAV	4
Minimum missile launch range (km)	5
Maximum missile launch range (km)	64
Explosion radius (km)	0.3
Maximum mid-course guidance time (s)	60
Maximum speed of missile (m/s)	1360
Minimum speed of missile (m/s)	680
Initial proportional guidance coefficient	6
Maximum guidance angle behind the tail of the missile	$\pm 50\pi/180$
Maximum missile's radar search range (km)	20
Maximum missile's radar search angle (rad)	$\pm 25\pi/180$

Table 4 Hyperparameters configurations.

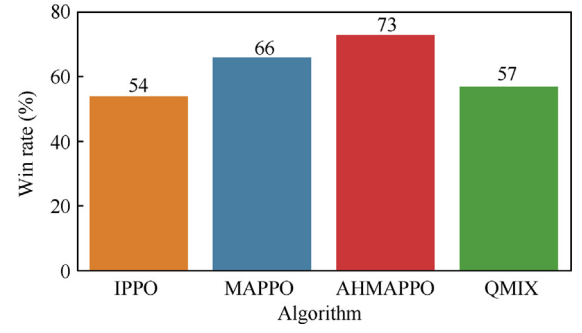
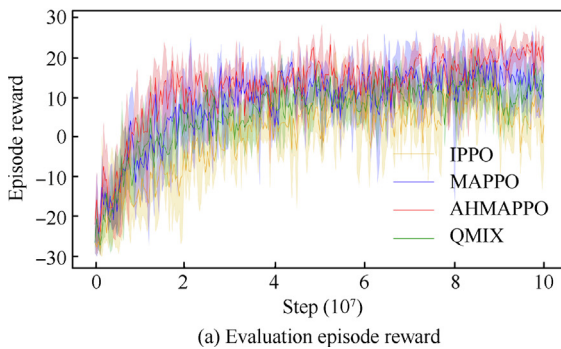
Parameter	Value
Parallel environments	5 (1 for QMIX)
Episode length	3000
No. of MLP layer	2
MLP layer dim	128
No. of RNN layer	1
RNN hidden state dim	128
No. of fully connected layer after	1
Recurrent data chunk length	1000
GAE lamda	0.95
GAE gamma	0.99
PPO epoch	5
No. of mini-batch	1
Learning rate	5×10^{-4}
Activation	ReLU
Clip parameter	0.2
Regulatory factor for the advantage highlight	0.01

In the UCAV formation BVR air combat simulation environment, the air combat space is a three-dimensional cylindrical area with a height range of $[0.5, 18]$ km and a cylindrical diameter of 160 km. The horizontal positions of red-side and blue-side UCAVs are randomly initialized within their respective semicircle with an altitude range of $[0.4 \times 18, 0.6 \times 18]$ km, pitch and roll angle range of $[-1^\circ, 1^\circ]$, and yaw angle range of $[-20^\circ, 20^\circ]$. All of the UCAVs' speeds are set at 400 m/s and remain unchanged. The rest of the confrontation details have been introduced in Section 2 and will not be repeated here. The environment parameters settings are shown in Table 3.

The confrontation of 3vs.3 cooperative BVR air combat is carried out in five random seeds. For each random seed, this paper evaluates the current strategy by computing the win rate over 10 evaluation environments with different seeds every 25 training episodes and taking the median of the final 5 evaluation win rates as the performance for the algorithm in each seed. The algorithms' main hyperparameters are configured in Table 4.

The convergence curves of the average evaluation episode reward and evaluation win rate are obtained, as shown in Fig. 8. The curves in Fig. 8(a) fluctuate more than the curves in Fig. 8(b) because the same win rate corresponds to multiple ways to win with different rewards. The final performance is reported in Fig. 9.

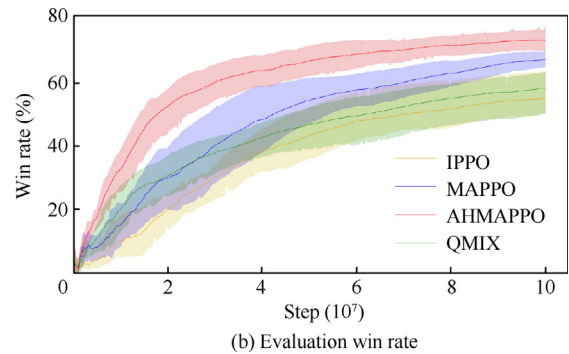
It can be seen from Fig. 8 that compared with other tested algorithms, the AHMAPPO proposed in this paper can con-

**Fig. 9** Final performance of algorithms.

verge faster and get a strategy with a higher win rate and episode reward. To be specific, the IPPO, even without the global information in the training process, still has an excellent effect on some multi-agent benchmarks.³⁹ However, when faced with the more complex UCAV formation BVR air combat simulation environment, the IPPO performs relatively terribly. Moreover, the MAPPO and QMIX with CTDE architecture have more outstanding convergence effects than the IPPO. But in the optimization process of the loss function, the MAPPO and QMIX tend to ignore the advantages of the current strategy in some cases because they only take the average performance in batch data for updating. By contrast, the AHMAPPO highlights the potential advantage of the current strategy. Consequently, it achieves faster convergence and obtains the best strategy among the tested algorithms with a win rate of 73% (Fig. 9), which has 35.2%, 10.6%, and 28.1% improvement over the IPPO, MAPPO, and QMIX, respectively. Especially in the early and middle stages of training, the advantage highlight method makes the AHMAPPO more sensitive to the optimal strategy so that the AHMAPPO can converge more efficiently.

To demonstrate the optimization efficiency of the AHMAPPO more clearly, this paper compares the sample episodes utilized by these algorithms to converge to the strategy with the same win rate in Fig. 10. By calculating the ratio of the difference between the episodes used by AHMAPPO and other algorithms to the episodes used by other algorithms under the same win rate, the sample efficiency improvement of AHMAPPO can be obtained in Table 5.

As shown in Fig. 10, the AHMAPPO can use fewer sample episodes than the other three algorithms to converge to a strategy with a certain win rate. When the number of sample episodes is greater than 21130, the AHMAPPO can obtain the

**Fig. 8** Convergence curves of algorithms in the UCAV formation BVR air combat simulation environment.

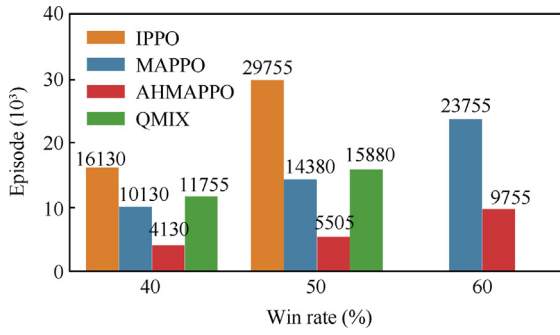


Fig. 10 Sample episodes utilized by different algorithms.

Table 5 Sample efficiency improvement of AHMAPPO relative to other algorithms at different win rates.

Win rate (%)	Sample efficiency improvement (%)		
	IPPO	MAPPO	QMIX
40.0	74.4	59.2	64.9
50.0	81.5	61.7	65.3
60.0	-	58.9	-

strategy with a win rate of no less than 70%, which is not displayed in Fig. 10 because no other algorithm can converge to a strategy with the same win rate. It is evident from Fig. 10 that the AHMAPPO has a more effective optimization ability. For example, when the number of sample episodes is about 9755, the AHMAPPO can converge to the strategy that has a certain probability of defeating the opponent, while the other algorithms can just obtain a strategy with a win rate of less than 40%. In Table 5, it can be seen more clearly that the sample efficiency improvement of AHMAPPO relative to the IPPO, MAPPO, and QMIX is pretty significant, and its maximum is 81.5%. Considering that one episode equals 3000 steps in the UCAV formation BVR air combat simulation environ-

ment, the improved convergence efficiency of AHMAPPO means that lots of convergence time is saved, which is significant for pre-war strategy preparation and wartime strategy iterative optimization.

From the results, it can be concluded that the AHMAPPO can converge to a more excellent strategy for the UCAV formation in a shorter time compared with the other three state-of-the-art algorithms in the UCAV formation BVR air combat simulation environment. In addition, we also show the performance of AHMAPPO in larger-scale confrontation scenarios, and details can be seen in the Appendix A.

5. Strategy analysis

In order to better show the effect of the 3vs.3 cooperative confrontation strategy obtained by the AHMAPPO, three typical strategies are summarized by replaying the cooperative confrontation process in the simulation environment. Combined with the specific combat rounds, three typical strategies are analyzed and explained as follows:

(1) Pincer offensive strategy

This strategy allows the red-side formation to attract the attention of the enemy by actively exposing two UCAVs in the formation while the remaining one keeps silent and approaches the enemy quickly. Until the conditions for radar search and missile launch are met, the radar of the UCAV, which was silent before, is suddenly turned on for missile launch and midcourse guidance. At this time, the roles of the UCAVs in the formation have changed; that is, the UCAV, which previously served as silent assault, is engaged in continuous mid-course guidance while attracting the attention of the enemy. The UCAVs, which previously served as fire attractions, switch to launching missiles from the flank to attack the enemy stealthily. During the entire BVR air combat, the formation is divided into two parts: sharing the enemy information through the data link and carrying out the mid-course guidance collaboratively. The two parts' UCAVs sup-

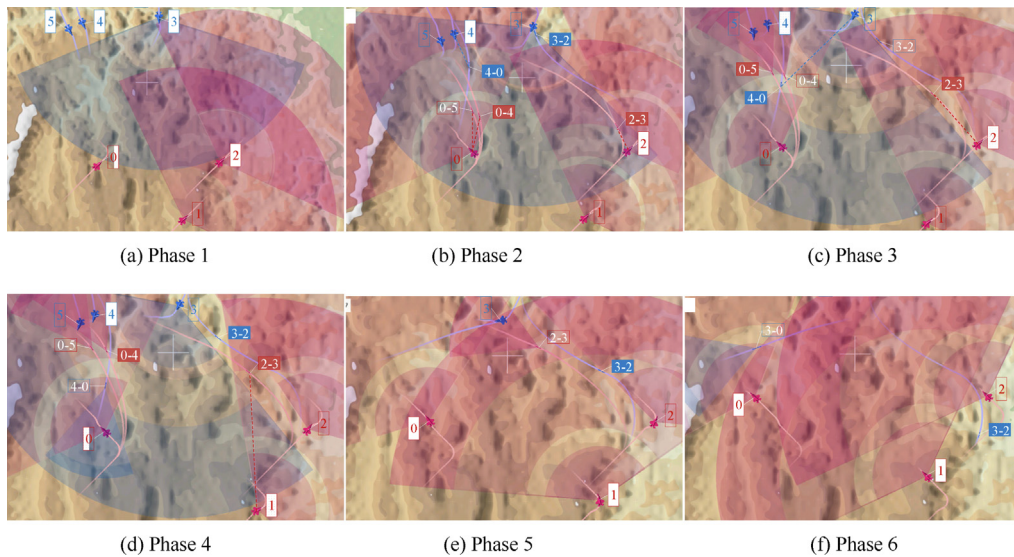


Fig. 11 Pincer offensive strategies.

port each other to form a pincer offensive to complete the attack, as seen in Fig. 11.

In Fig. 11(b), the pincer offensive strategy makes the red-side formation obtain the accurate information of all the blue-side UCAVs. So the red-side formation realizes the first view and first shots the Missile 0-5 (refers to the missile launched by the UCAV 0 towards the UCAV 5) and 2-3. Additionally, some guidance modes emerge in the mid-course guidance phase. The UCAV 4 provides guidance for the Missile 4-0 while the UCAV 3 is tracking the target in Fig. 11(b), that is, the guidance mode of “One UCAV launches and guides, the other searches”. The UCAV 1 and the UCAV 2 adopt the guidance mode of “Collaborative guidance by ESM” to guide the Missile 2-3 in Fig. 11(b). The UCAV 3 takes over the guidance of the Missile 4-0 from the UCAV 4 in Fig. 11(c), which is the guidance mode of “Alternate guidance”. Finally, after receiving adequate mid-course guidance, the red-side formation realizes the first kill in Fig. 11(f) and wins.

(2) Outflanking strategy

As shown in Fig. 12, this strategy enables the red-side formation to separate two UCAVs to keep silent and flank the battlefield. Meanwhile, another UCAV attracts the enemy's fire in the front and turns on the radar to obtain the enemy information as much as possible. In light of the shared information, the flanking UCAVs launch missiles at the target and guide the missiles in the mid-course guidance phase, causing the enemy to be attacked from the flank while dealing with a frontal threat.

The Outflanking strategy makes the red-side formation achieve the first view and first shot of the Missile 0-5 in Fig. 12(c). What's more, in Fig. 12(c), the UCAV 3 tracks the target and guides the Missile 4-2 launched by UCAV 4 utilizing the guidance mode of “One UCAV launches, the other searches and guides”. In the end, the red-side formation achieves the first kill in Fig. 12(f) and wins.

(3) Thorough search strategy

In Fig. 13, this strategy makes all UCAVs in the red-side formation keep the radar on and maintain triangular formation. The most forward UCAV searches the front battlefield continuously, and the rear UCAVs keep controlling the battlefield situation on the flanks of the red-side formation. Once there is a target within the search range, based on their own large area control of the battlefield situation, the red-side formation can realize a more efficient attack on the target than the enemy through collaborative guidance.

The thorough search strategy enables the red-side formation to realize the first view in Fig. 13(a) and the first shot of the Missile 0-5 in Fig. 13(b). The UCAV 3 tracks the target and guides the Missile 4-1 in Fig. 13(e), that is, the guidance mode of “One UCAV launches, the other searches and guides”. The UCAV 1 and UCAV 2 guide the Missile 1-3 with the target's information obtained by ESM in Fig. 13(g), that is the guidance mode of “Collaborative guidance by ESM”. Finally, the red-side formation achieves the first kill in Fig. 13(i) and wins.

It should be pointed out that in addition to the above three strategies, the 3vs.3 cooperative confrontation strategy obtained by the AHMAPPO also contains a variety of detailed strategies, including evasion strategies for threatening missiles, tracking strategies for establishing communication with missiles or continuous detection of targets, and selection of radar status under different situations, etc. All of those can be reflected in the three above summarized strategies. It can be concluded that the AHMAPPO proposed in this paper can converge to the strategy, which utilizes radar, ESM, data link, and maneuvers intelligently to win the BVR air combat cooperatively.

6. Conclusions

This paper studies the cooperative decision-making algorithm for the UCAV formation in BVR air combat based on MARL.

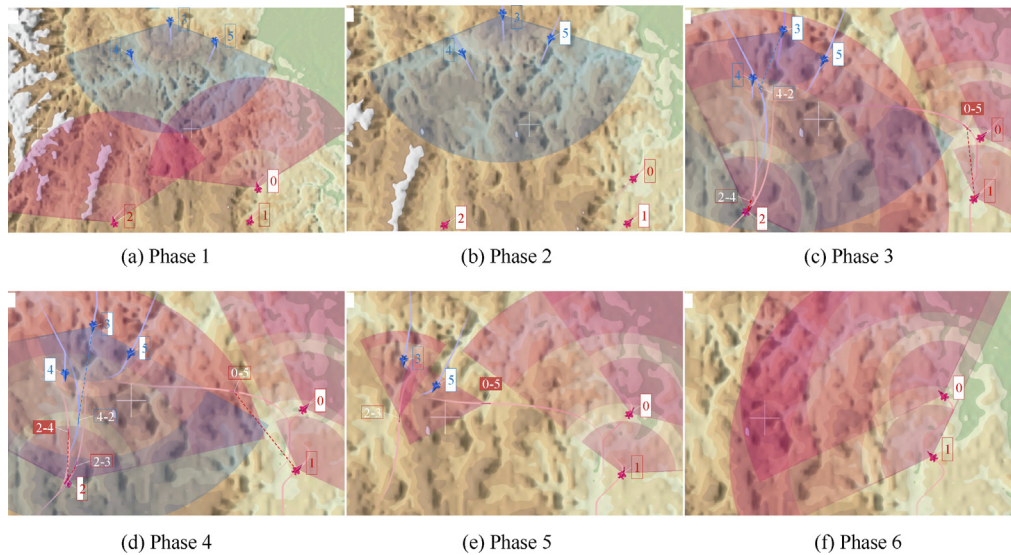


Fig. 12 Outflanking strategies.

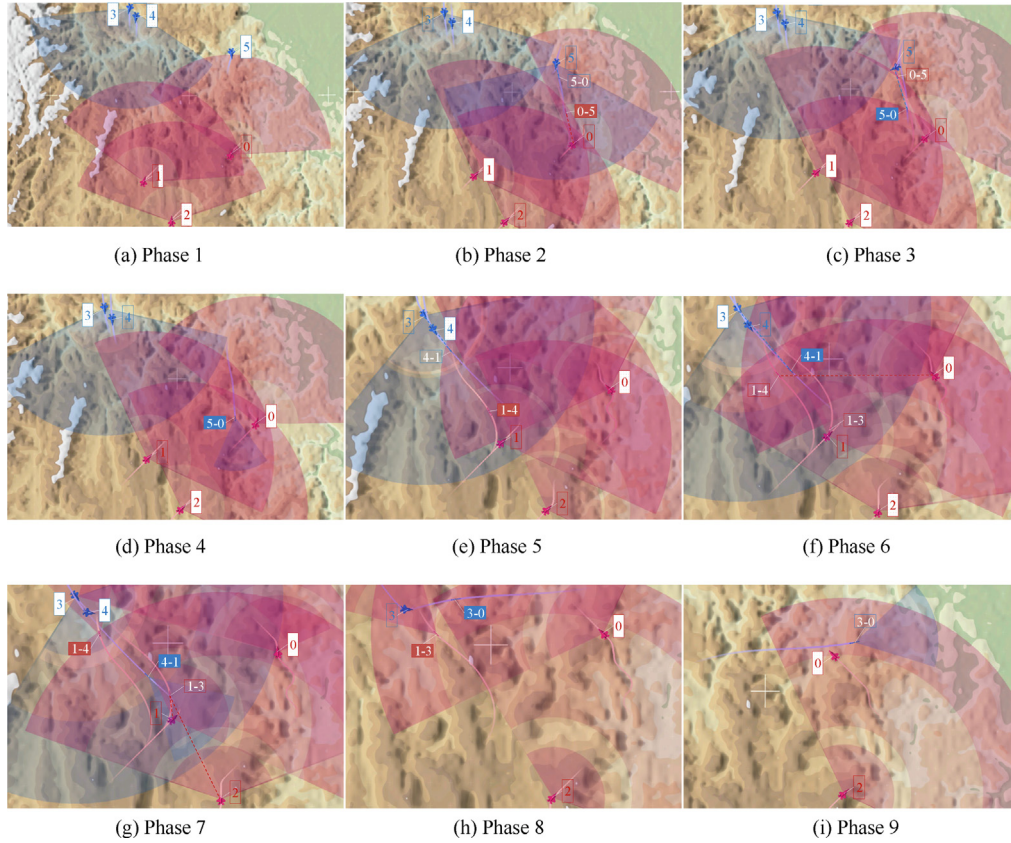


Fig. 13 Thorough search strategies.

According to the basic characteristics of the BVR air combat, this paper builds the UCAV formation BVR air combat simulation environment involving the partially observable battlefield, communication between UCAVs based on the data link, and mid-course guidance phase of the missile to expose the low strategy convergence efficiency problem. To address the above problem that MARL algorithms have in the UCAV formation BVR air combat simulation environment, an AHMAPPO algorithm is proposed. To be specific, while optimizing the actor network, extra emphasis is placed on the degree to which the best formation exceeds the average of formations in parallel environments to improve the convergence speed and effect of the actor network so that more excellent strategies can be converged in a shorter time. The complex cooperative confrontation strategy for the blue-side formation is designed to support the training of the red-side agents on the basis of the AHMAPPO.

By utilizing the AHMAPPO, the 3vs.3 cooperative confrontation strategy with a win rate of 73%, which has 35.2%, 10.6%, and 28.1% improvement over the IPPO, MAPPO, and QMIX, respectively, is obtained in the UCAV formation BVR air combat simulation environment. The sample efficiency improvement of AHMAPPO relative to the above algorithms is also significant, and its maximum is 81.5%. Intended to show the effect of the strategy, three typical strategies are analyzed and explained in this paper. All the above results demonstrate that the AHMAPPO proposed in this paper is meaningful for solving the convergence problem of cooperative confrontation strategy in BVR air combat.

In future studies, considering the need for systematic confrontation, more research will be conducted to investigate the cooperative decision-making problems for the heterogeneous UCAV formation in BVR air combat.

CRediT authorship contribution statement

Yaoming ZHOU: Funding acquisition, Project administration, Resources, Supervision. **Fan YANG:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Chaoyue ZHANG:** Investigation, Visualization. **Shida LI:** Investigation. **Yongchao WANG:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Simulation results of AHMAPPO in larger-scale confrontation scenarios

The confrontation of 5vs.5 cooperative BVR air combat is carried out in three random seeds. It should be noted that due to the increase in the number of UCAVs, the dimension of local observation and global state will expand, especially the global state that contains more information about the simulation environment. To ensure the data processing capability of the critic network, the MLP layer dim and RNN hidden state dim of the critic network in Table 4 are changed to 256 in the confrontation scenario of 5vs.5. Moreover, We have also tested the algorithms' performance in the confrontation scenario of 7vs.7, in which, the MLP layer dim and RNN hidden state dim of the actor network in Table 4 are also changed to 256. The convergence curves of the average evaluation episode reward and evaluation win rate in the confrontation scenario of 5vs.5 and 7vs.7 are obtained, as shown in Fig. A1 and Fig. A2, respectively.

Fig. A1 and Fig. A2 show that compared with other tested algorithms, the AHMAPPO can still converge faster to a strategy with a higher win rate and episode reward in the confrontation scenario of 5vs.5 and 7vs.7. However, compared with the confrontation scenario of 3vs.3, due to the expansion of the information dimension input to the actor and critic networks, the information processing difficulty of the neural network is increased, which reduces the convergence efficiency of the MARL algorithm, and the win rate of the optimized strategy also decreases overall, as shown in Fig. A3 and Fig. A4. This reflects the inherent scalability problem of the MARL algorithm; that is, with the expansion of the number of agents and the dimension of the information, the algorithm's convergence efficiency and effect are difficult to guarantee.

To demonstrate the optimization efficiency of the AHMAPPO in the confrontation scenario of 5vs.5 and 7vs.7 more clearly, we also compare the sample episodes utilized by these algorithms to converge to the strategy with the same win rate in Fig. A5 and Fig. A6. The sample efficiency improvement of AHMAPPO can be obtained in Table A1.

Fig. A1 Convergence curves of algorithms in the confrontation scenario of 5vs.5.

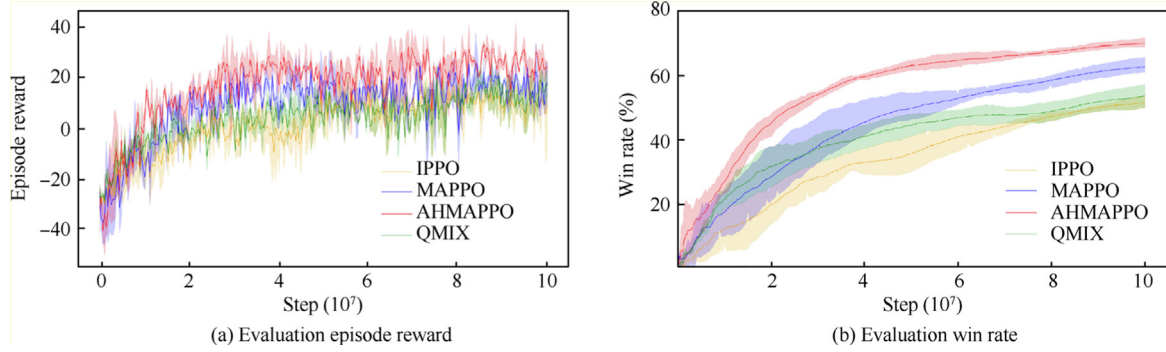


Fig. A2 Convergence curves of algorithms in the confrontation scenario of 7vs.7 (because of the large scale of data processing and the long training time, only the algorithms' performance in one random seed is provided here).

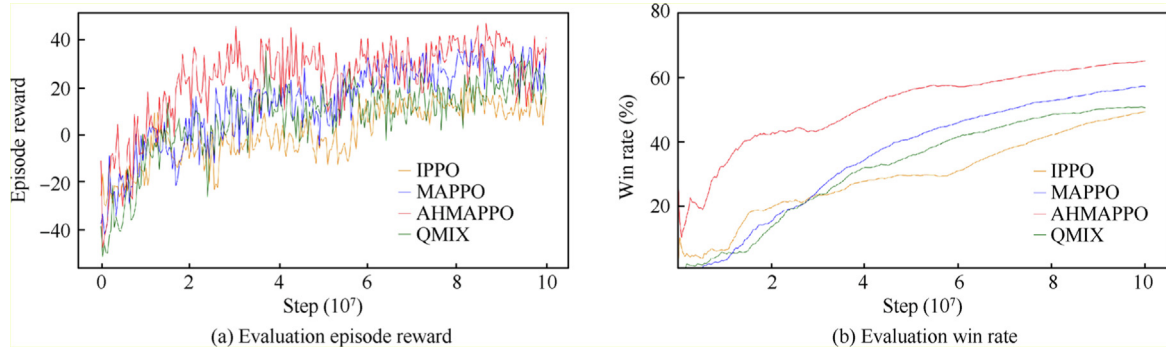
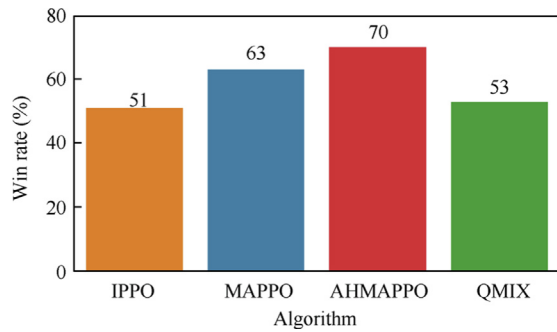
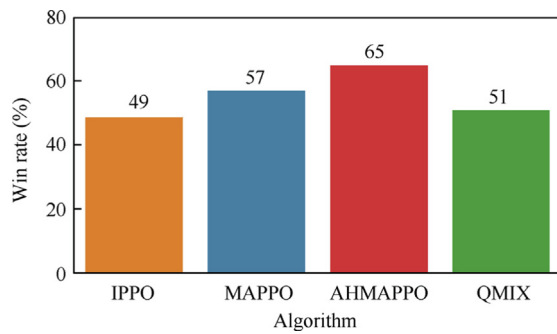
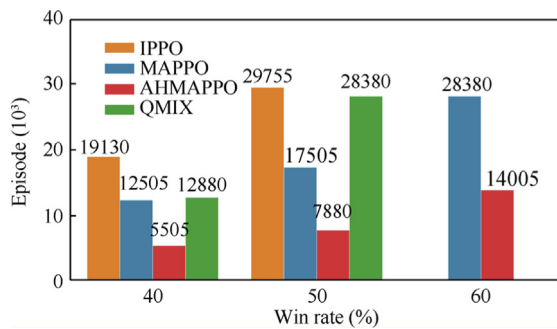
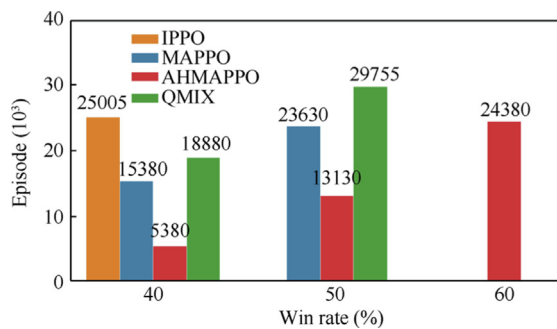


Fig. A3 Final performance of algorithms in 5vs.5.**Fig. A4** Final performance of algorithms in 7vs.7.**Fig. A5** Sample episodes utilized in 5vs.5.**Fig. A6** Sample episodes utilized in 7vs.7.**Table A1** Sample efficiency improvement of AHMAPPO relative to other algorithms at different win rates.

Scenario	Win rate (%)	Sample efficiency improvement (%)		
		IPPO	MAPPO	QMIX
3vs.3	40.0	74.4	59.2	64.9
	50.0	81.5	61.7	65.3
	60.0	-	58.9	-
5vs.5	40.0	71.2	56.0	57.3
	50.0	73.5	55.0	72.2
	60.0	-	50.7	-
7vs.7	40.0	78.5	65.0	71.5
	50.0	-	44.4	55.9
	60.0	-	-	-

Compared with the 3vs.3 confrontation scenario, although the above two scenarios do not increase too many UCAVs, due to the complexity of the UCAV formation BVR air combat simulation environment constructed in this paper, the 5vs.5 and 7vs.7 confrontation scenarios have a considerable expansion in the dimension of global state and local observation, which has put forward a challenge to the convergence of the MARL algorithm. Therefore, the performance of all four algorithms deteriorates in the face of larger confrontation scenarios. Nevertheless, It can be seen from Fig. A5, Fig. A6, and Table A1 that the AHMAPPO can use fewer sample episodes than the other three algorithms to converge to a strategy with a certain win rate in the confrontation scenario of 5vs.5 and 7vs.7. Moreover, because of the expansion of the strategy space brought about by the increase in the number of agents, the compensation effect of A^* has indeed been affected. However, the AHMAPPO can still maintain a specific convergence efficiency improvement ability compared with the other three state-of-the-art MARL algorithms in the same confrontation scenario.

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